The Kiran Academy | IT Training & Placement

Python Variables Practice Assignment

50 Interview-Oriented Questions | Beginner → Intermediate → Advanced

Instructions: Write a separate Python program for each task. First solve independently, then explain your approach as if you are in an MNC interview. Use basic Python concepts wherever possible.

Section A: Beginner Level — Variable Fundamentals

1. Create variables **name**, **age**, **city**, and **course**. Print their values.
2. Store your first name and last name in two variables and print the full name.
3. Create variables for product name, price, and quantity. Calculate and print the total bill.
4. Store two numbers in variables and print their addition, subtraction, multiplication, and division.
5. Create a variable and use **type()** to display its data type.
6. Store your age in a variable and calculate your approximate birth year.
7. Store temperature in Celsius and convert it to Fahrenheit using variables.
8. Store marks of three subjects in variables and calculate total and average.
9. Swap the values of two variables without using a third variable.
10. Take a user's name and age using **input()**, store them in variables, and display a welcome message.

Section B: Beginner Story-Based Practice

11. **Student Admission:** Store student name, age, course, and fees in variables and display an admission summary.
12. **Shopping Cart:** Store product name, price, and quantity. Calculate the total amount.
13. **Employee Salary:** Store basic salary, HRA, and allowance. Calculate gross salary.
14. **Restaurant Bill:** Store item price, quantity, and discount percentage. Calculate the final bill.
15. **Exam Result:** Store marks for five subjects and calculate total, average, and percentage.
16. **Bank Account:** Store account holder name and balance. Perform a deposit and withdrawal using variables.
17. **Travel Booking:** Store passenger name, destination, ticket price, and number of passengers. Calculate total cost.
18. **Electricity Bill:** Store units consumed and price per unit. Calculate the bill amount.
19. **Gym Membership:** Store member name, monthly fee, and number of months. Calculate total membership cost.
20. **Mobile Recharge:** Store customer name, recharge amount, and validity. Print a recharge confirmation.

Section C: Intermediate Level — Logic with Variables

21. Take two numbers from the user and store them in variables. Display which number is larger.
22. Store a number in a variable and determine whether it is positive, negative, or zero.
23. Store student marks and calculate the grade using **if/elif/else**.
24. Store salary and years of experience. Calculate a bonus based on experience.
25. Store principal, rate, and time and calculate simple interest.

26. Store cost price and selling price. Determine whether the transaction gives profit or loss.
27. Store three numbers and find the largest number without using **max()**.
28. Store a year in a variable and determine whether it is a leap year.
29. Store hours worked and hourly rate. Calculate regular pay and overtime pay.
30. Store purchase amount and membership status. Apply different discounts using conditions.

Section D: Intermediate/Advanced Story-Based MNC Scenarios

31. **E-commerce Checkout:** Store product price, quantity, coupon status, and membership status. Calculate the final payable amount.
32. **Employee Performance:** Store sales amount, attendance percentage, and performance rating. Decide whether the employee receives an incentive.
33. **Loan Eligibility:** Store age, monthly income, and credit score. Create a basic eligibility rule.
34. **Cab Fare:** Store distance, base fare, per-kilometre rate, and peak-hour status. Calculate the final fare.
35. **Online Exam:** Store correct answers, attempted answers, marks per question, and negative marking. Calculate the score.
36. **Inventory:** Store product name, available stock, reorder level, and requested quantity. Decide whether the order can be fulfilled.
37. **Attendance:** Store total working days and present days. Calculate attendance percentage and determine eligibility for a bonus.
38. **Sales Analysis:** Store total sales, number of orders, and target sales. Calculate average order value and target achievement.
39. **System Monitoring:** Store CPU usage, memory usage, response time, and error count. Classify the system as healthy, warning, or critical.
40. **Payroll:** Store basic salary, overtime hours, overtime rate, deductions, and tax rate. Calculate gross and net salary.

Section E: Advanced Interview & Conceptual Practice

41. Demonstrate the difference between **=**, **==**, and **is** using variables.
42. Create a program demonstrating variable reassignment and explain what happens when a variable receives a new value.
43. Demonstrate dynamic typing by assigning different data types to the same variable at different times.
44. Demonstrate local and global variables using a function. Explain the scope of each variable.
45. Demonstrate variable shadowing using a global variable and a local variable with the same name.
46. Create two variables that reference the same list. Modify the list through one variable and explain the result.
47. Use **id()** to investigate whether two variables refer to the same object.
48. Demonstrate multiple assignment, chained assignment, and variable unpacking.
49. Debug a program where **input()** values are used directly in arithmetic. Explain why conversion is required.
50. **MNC Interview Challenge:** A production program contains variables named **x, y, z, a, b** for salary, tax, sales, employee ID, and department. Refactor the variables using meaningful names and explain how good naming improves readability, maintenance, and debugging.